FSM Analysis

Contemporary Digital Electronic Systems — Chapter 14

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Why This Chapter Matters

Design goes from specification to circuit; analysis goes the other way. Given a sequential schematic, can you recover the state diagram and the behaviour? That is how you document legacy circuits, debug someone else's design, and verify your own.

After this chapter, you can:

1. Identify the state register and the next-state and output logic in a schematic.
2. Derive the flip-flop input and output equations by inspection.
3. Build the state table from those equations.
4. Draw the state diagram from the state table.
5. Classify the recovered FSM as Moore or Mealy.

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If you were handed an unlabelled chip containing a small FSM, what would you do to reconstruct its state diagram?

This chapter does it from the schematic, which is the same job on paper.

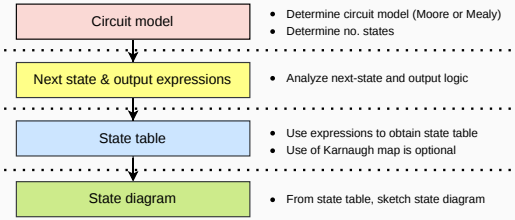
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Reverse Engineering an FSM

- Analysis is reverse engineering: understand a system from its structure and operation.
- Needed when a legacy circuit must be rebuilt (parts discontinued, documentation lost), to understand undocumented designs, to estimate cost and find improvements, even to assess patent infringement.
- Reverse engineering is limited by law and contract. Analysing a circuit to understand or repair it is generally defensible; copying a protected design is not.



1. Identify the model (Moore or Mealy) and the number of states.
2. Derive the next-state and output expressions.
3. Build the state table (K-maps optional).
4. Draw the state diagram.

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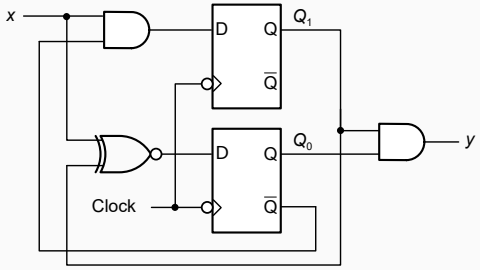
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Moore Circuit: Steps 1 and 2



Step 1: model and number of states

Output y comes only from flip-flop outputs, with no direct path from x: Moore. Two flip-flops: up to 4 states.

Step 2: equations

$D_1 = Q_0'x \qquad D_0 = \overline{Q_1 \oplus x} = Q_1x + Q_1'x'$

D flip-flops: $Q^+ = D$, so these are the next-state equations.

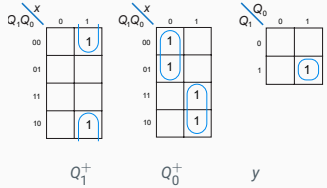
$y = Q_1Q_0$

Moore Circuit: Minterms

Canonical SOP route (indices Q_1Q_0x):

$$Q_1^+ = Q_0'x = Q_1'Q_0'x + Q_1Q_0'x = \sum m(1,5)$$

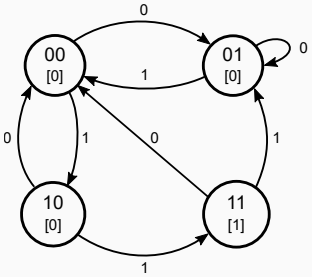
$$Q_0^+ = Q_1x + Q_1'x' = \sum m(0,2,5,7)$$



- Either the K-map or the canonical-SOP route gives the same state table.

Moore Circuit: State Table and Diagram

Q_1Q_0	x	$Q_1^+Q_0^+$	y
00	0	01	0
00	1	10	0
01	0	01	0
01	1	00	0
10	0	00	0
10	1	11	0
11	0	00	1
11	1	01	1



- Step 4: each table row becomes an arrow; the output sits inside each state.

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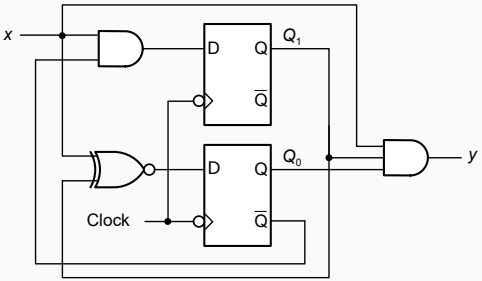
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Mealy Circuit

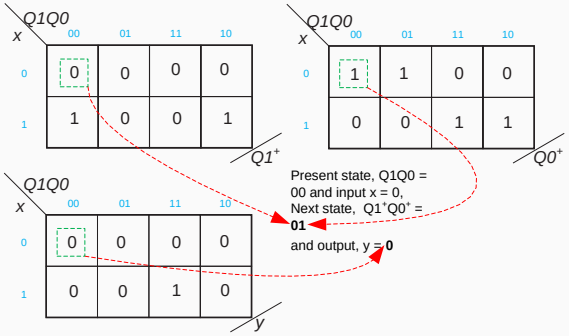


Steps 1 and 2

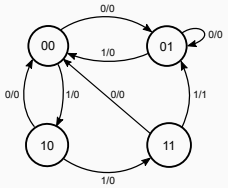
The input x goes straight to the rightmost AND gate: Mealy.
Two flip-flops: $2^2 = 4$ states.

$Q_1^+ = D_1 = Q_0' x$
$Q_0^+ = D_0 = Q_1 x + Q_1' x'$
$y = Q_1 Q_0 x$

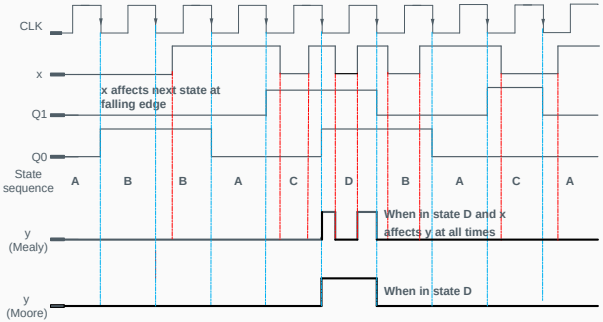
Mealy Circuit: State Table from a K-Map



- Lay out a K-map whose inputs are the present state Q_1Q_0 and x.
- Fill each cell with the next state $Q_1^+Q_0^+$ and output y.
- Each cell is one arrow of the state diagram.



Moore versus Mealy Waveforms



- A waveform can replace a state diagram for showing next state and output.
- The Moore output changes synchronously with the clock edge.
- The Mealy output can glitch: the asynchronous input *x* changes it immediately, whatever the clock does.
- These waveforms sample on the falling edge; the rest of the book uses the rising edge. Edge sensitivity is fixed by the flip-flops, so check the schematic.

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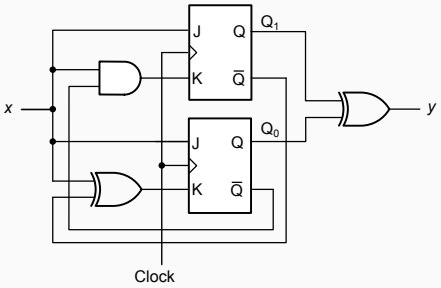
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Analysing a JK-Based Circuit



- JK flip-flops are not recommended for new designs, but legacy TTL circuits still use them.
- Step 1: the output depends only on the state: Moore; $2^2 = 4$ states.
- Step 2: flip-flop inputs $J_1 = x, K_1 = Q_0'x, J_0 = x, K_0 = Q_1' \oplus x$; output $y = Q_1 \oplus Q_0$.

JK Circuit: Next-State Equations

- Substitute into the JK characteristic equation $Q^+ = JQ' + K'Q$.

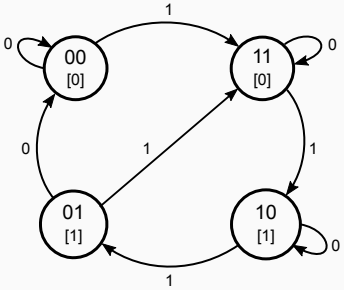
$$Q_1^+ = xQ_1' + (Q_0'x)'Q_1 = Q_1'x + Q_1Q_0 + Q_1x' = \sum m(1,3,4,6,7)$$
$$Q_0^+ = xQ_0' + (Q_1' \oplus x)'Q_0 = Q_0'x + Q_1'Q_0x + Q_1Q_0x' = \sum m(1,3,5,6)$$

Unlike the D flip-flop

For a JK flip-flop the input equations are *not* the next-state equations: always apply the characteristic equation.

JK Circuit: State Table and Diagram

Q_1Q_0	x	$Q_1^+Q_0^+$	y
00	0	00	0
00	1	11	0
01	0	00	1
01	1	11	1
10	0	10	1
10	1	01	1
11	0	11	0
11	1	10	0



Reconstructing the state diagram

For a real chip you would probe the flip-flop outputs and the output pins while applying input patterns, and record, for each present state and input, the next state and output: you would be building the state table directly. This chapter does the same on paper, with the equations read off the schematic, so the table can be filled in exhaustively: the same procedure, driven by measurement instead of a schematic.

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Summary

- D flip-flop circuits are the easiest to analyse: $Q^+ = D$, so the input equations are also the next-state equations.
- For JK flip-flops, use the characteristic equation to convert input equations into next-state equations.
- Enumerate every (state, input) pair to fill the state table; read the state diagram from it.
- A direct path from an input to an output means Mealy; otherwise Moore.

Looking ahead

Chapter 15 scales the design procedure up to controllers: FSMs with several inputs and outputs that drive the rest of a system.

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